

A Laboratory Manual for

Web Programming
(3160713)

B.E. Semester 6
(Computer Engineering)



Directorate of Technical Education, Gandhinagar, Gujarat

GOVERNMENT ENGINEERING COLLEGE DAHOD

Affiliated to GUJARAT TECHNOLOGICAL UNIVERSITY

COMPUTER ENGINEERING DEPARTMENT



CERTIFICATE

This is to certify that **Shaikh Ajim** bearing enrollment number **240183107021** has satisfactorily completed the term work under the program **B.E. Computer Engineering** of **6th** semester course **Web Programming (3160713)** for the academic term **EVEN 2025 – 26** ending in **May 2026**.

Date of Submission

___/___/___

Sign of Faculty

DTE's Vision

- To provide globally competitive technical education
- Remove geographical imbalances and inconsistencies
- Develop student friendly resources with a special focus on girls' education and support to weaker sections
- Develop programs relevant to industry and create a vibrant pool of technical professionals

Institute's Vision

To be a value based engineering institute to disseminate globally acceptable education and to nurture research, innovation and entrepreneurship.

Institute's Mission

- To provide quality education in engineering disciplines through creative balance of academic and extra-curricular activities.
- To provide learning environment for innovation and entrepreneurship.
- To disseminate ethical values, social values and sensitivity towards environmental issues.

Department's Vision

- To be a distinguished department that provides an environment conducive to learning, innovation and research while nurturing an inclusive and all-round development of its stakeholders.

Department's Mission

- To prepare computer engineers capable of implementing professionally valued solutions.
- To imbibe aspects of lifelong learning, adapting technological change and addressing needs of society by keeping abreast of advancements in engineering and technology.

Preface

Main motto of any laboratory/practical/field work is for enhancing required skills as well as creating ability amongst students to solve real time problem by developing relevant competencies in psychomotor domain. By keeping in view, GTU has designed competency focused outcome-based curriculum for engineering degree programs where sufficient weightage is given to practical work. It shows importance of enhancement of skills amongst the students and it pays attention to utilize every second of time allotted for practical amongst students, instructors and faculty members to achieve relevant outcomes by performing the experiments rather than having merely study type experiments. It is must for effective implementation of competency focused outcome-based curriculum that every practical is keenly designed to serve as a tool to develop and enhance relevant competency required by the various industry among every student. These psychomotor skills are very difficult to develop through traditional chalk and board content delivery method in the classroom. Accordingly, this lab manual is designed to focus on the industry defined relevant outcomes, rather than old practice of conducting practical to prove concept and theory.

By using this lab manual students can go through the relevant theory and procedure in advance before the actual performance which creates an interest and students can have basic idea prior to performance. This in turn enhances pre-determined outcomes amongst students. Each experiment in this manual begins with competency, industry relevant skills, course outcomes as well as practical outcomes (objectives). The students will also achieve safety and necessary precautions to be taken while performing practical.

This manual also provides guidelines to faculty members to facilitate student centric lab activities through each experiment by arranging and managing necessary resources in order that the students follow the procedures with required safety and necessary precautions to achieve the outcomes. It also gives an idea that how students will be assessed by providing rubrics.

In the era of digitization, the demand of Internet based applications is increasing day by day. Web Development is one of the required skills for IT Engineer. This focuses on front-end and back-end design. After learning Web Development students can advance their career in the field of web development.

Utmost care has been taken while preparing this lab manual however always there is chances of improvement. Therefore, we welcome constructive suggestions for improvement and removal of errors if any.

Industry Relevant Skills

The following industry relevant competency are expected to be developed in the student by undertaking the practical work of this laboratory.

1. **HTML/CSS Skills:** HTML is used extensively by web developers to build web pages. CSS is used to implement different fonts, colors and layouts in the design of a website.
2. **Javascript Skills:** Java Script is used for creating interactive web pages to improve the user experience.
3. **PHP/MySQL Skills:** PHP/MySQL is used extensively by web developers to create fully functional dynamic web applications.
4. **REST API, AJAX, JQuery Skills:** REST API, AJAX, JQuery are advanced asynchronous web communication mechanisms and used by web developers for building highly interactive webpages.

Guidelines for Faculty members

1. Teacher should provide the guideline with demonstration of practical to the students with all features.
2. Teacher shall explain basic concepts/theory related to the experiment to the students before starting of each practical
3. Involve all the students in performance of each experiment.
4. Teacher is expected to share the skills and competencies to be developed in the students and ensure that the respective skills and competencies are developed in the students after the completion of the experimentation.
5. Teachers should give opportunity to students for hands-on experience after the demonstration.
6. Teacher may provide additional knowledge and skills to the students even though not covered in the manual but are expected from the students by concerned industry.
7. Give practical assignment and assess the performance of students based on task assigned to check whether it is as per the instructions or not.
8. Teacher is expected to refer complete curriculum of the course and follow the guidelines for implementation.

Instructions for Students

1. Students have to write answers / solutions of QUIZ in separate file page. The quiz of corresponding practical must be attached just behind each practical.
2. Students are expected to carefully listen to all the theory classes delivered by the faculty members and understand the COs, content of the course, teaching and examination scheme, skill set to be developed etc.
3. Students shall organize the work in the group and make record of all observations.
4. Students shall develop maintenance skill as expected by industries.
5. Student shall attempt to develop related hand-on skills and build confidence.
6. Student shall develop the habits of evolving more ideas, innovations, skills etc. apart from those included in scope of manual.
7. Student shall refer technical magazines and data books.
8. Student should develop a habit of submitting the experimentation work as per the schedule and s/he should be well prepared for the same.

Index (Progressive Assessment Sheet)

Sr. No.	Objective(s) of Experiment	Page No.	Date of performance	Date of submission	Assessment Marks	Sign. of Teacher with date	Remarks
1.	Create a professional Resume using semantic HTML5 tags (header, footer, section, article, nav). Include sections for Personal Info, Education, Skills, and Hobbies using lists and appropriate text formatting.						
2.	Design a Class Timetable or Bus Schedule using the <table> tag. Strictly implement rowspan, colspan, cellspacing, and cellpadding to understand data structuring.						
3.	Create a "Student Event Registration" form. Utilize HTML5 input types (date, email, number, url, color), attributes (required, placeholder, pattern), and form elements like <select>, <textarea>, <fieldset>, and <legend>, etc.						
4.	Apply External and Internal CSS to the Resume created in Exp. 1. Experiment with the Box Model (margin, padding, border), Text formatting, google fonts, and Background properties.						
5.	Develop a "College Department Landing Page" using a CSS Framework like Bootstrap . The page must be responsive (mobile-first) and include a Navigation Bar, Jumbotron/Hero section, Grid system (Rows/Cols) for content, and a Footer.						
6.	Develop a JavaScript program to create a "Simple Calculator" and "EMI Calculator". Implement functions for arithmetic operations and event handling (onclick) to display results dynamically.						
7.	Create a "To-Do List" application. Users should be able to add tasks to a list, remove tasks, and mark them as complete using						
8.	Apply JavaScript validation to the Registration Form (from Exp. 3). Implement Regular Expressions (Regex) to validate Email format, Password strength, and Mobile number. Display custom error messages on the page.						
9.	Write a PHP script to generate a "Salary Slip". Take basic inputs (Basic Pay, DA%, HRA%) via an HTML form, process the data using PHP variables and arithmetic operators, and display the gross/net salary.						

10.	Implement a Login system. Store the username in a Session variable upon login to maintain state across pages. Implement a "Remember Me" feature using Cookies . Provide a Logout button to destroy the session.						
11.	Create a User Profile page that allows a user to upload a profile picture. Write a PHP script to validate the image type/size and save the file to the server directory.						
12.	Connect PHP with MySQL. Create a script to insert user details (from Exp. 3) into a database table. Implement errors handling database connection failures.						
13.	Develop a "Student Information Management System". Display records on an HTML table (Read). Add "Edit" and "Delete" buttons for each row. Handle the Update and Delete logic using PHP.						
14.	Create a "Live Search" feature. As the user types a name in a text box, use AJAX to fetch matching results from the MySQL database and display them immediately without reloading the page.						
15.	Develop a web page with a responsive image carousel (slider) using jQuery . Implement logic to transition images automatically every 3 seconds and provide "Next" and "Previous" navigation buttons to manually cycle through the gallery.						
16.	Create a simple REST API using PHP that accepts a GET request and returns a list of students (Enrolment No, Name and Branch) from the MySQL database and return the result as a strictly formatted JSON object to the client. Demonstrate how to consume this API using a browser or Postman.						
Total							